## Markov's Inequality

Let $u(X)$ be a nonnegative function of the random variable $X$. If $E(u(X))$, then for every positive constant $c$,

$$P(u(X) \geq c) \leq \frac{E(u(X))}{c}$$

### Proof

Let $A = \{x : u(x) \geq c\}$, $A^c = \{x : u(x) < c\}$

We assume $X$ is continuous (similar for discrete)

$$\Rightarrow E(u(X)) = \int_{S_x} u(x)f(x)dx = \underbrace{\int_A u(x)f(x)dx + \int_{A^c} u(x)f(x)dx}_{\text{both non-negative}}$$

$$\geq \int_A u(x)f(x)dx \geq \int_A cf(x)dx = cP(A) = cP(u(x) \geq c)$$

$\Rightarrow$ Now we have that

$$E(u(x)) \geq cP(u(x) \geq c) \Rightarrow \frac{E(u(x))}{c} \geq P(u(x) \geq c)$$

$$\Rightarrow P(u(X) \geq c) \leq \frac{E(u(X))}{c}$$

## Alternative Markov's Inequality

Let $X$ be a nonnegative random variable

$$P(X \geq c) \leq \frac{E(X)}{c}$$

### Proof

$$E(X) = \int_0^\infty xf(x)dx = \int_0^c xf(x)dx + \int_c^\infty xf(x)dx$$

$$\geq \int_c^\infty xf(x)dx$$

$$\geq \int_c^\infty cf(x)dx = c\int_c^\infty f(x) = cP(X \geq c)$$

$$E(X) \geq cP(X \geq c) \Rightarrow \frac{E(X)}{c} \geq P(X \geq c)$$

## Convex Functions

A function $\phi$ defined on an interval $(a,b)$ $-\infty \leq a < b \leq \infty$ is said to be a convex function if for all $x, y \in (a,b)$ and for all $0 < \gamma < 1$,

$$\phi[\gamma x + (1-\gamma)y] \leq \gamma\phi(x) + (1-\gamma)\phi(y)$$

If the inequality is strict, we say $\phi$ is strictly convex.

## Chebyshev's Inequality

Let $X$ be a random variable with finite variance $\sigma^2$, then for every $k > 0$

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Or equivalently,

$$P(|X - \mu| \leq k\sigma) > 1 - \frac{1}{k^2}$$

### Proof:

$$\Rightarrow P(|X-\mu| \geq k\sigma) = P((X-\mu)^2 \geq k^2\sigma^2) \leq \frac{E((X-\mu)^2)}{k^2\sigma^2} = \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}$$

$\Rightarrow$ Then we are left with:

$$\Rightarrow P(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$$

## Example

Let $X$ have pdf $f(x) = \frac{1}{2\sqrt{3}}$ for $-\sqrt{3} < x < \sqrt{3}$. Find $P(|X-\mu| \geq k\sigma)$ for $k = \frac{3}{2}$ and compare it to the upper bound provided by Chebyshev's inequality.

$$\mu = E(X) = \int_{-\sqrt{3}}^{\sqrt{3}} x \cdot \frac{1}{2\sqrt{3}} dx = 0$$

$$E(X^2) = \int_{-\sqrt{3}}^{\sqrt{3}} x^2 \cdot \frac{1}{2\sqrt{3}} = \sigma^2 = \mu^2 = 1$$

$$P(|X - 0| \geq k\sigma) = P(|X| \geq \tfrac{3}{2}) = 1 - \int_{-3/2}^{3/2} \frac{1}{2\sqrt{3}} dx = 1 - \frac{\sqrt{3}}{2} \approx 0.134$$

Chebyshev's Inequality guarantees an upper bound of $\frac{1}{k^2} = \frac{4}{9} \approx 0.444$